P1 may 2011 TZ1

1.

1. I is given as 0.10A using a digital meter so uncertainty = ±0.01A

% unc. = $\frac{0.01}{0.1} \times 100 = 10\%$

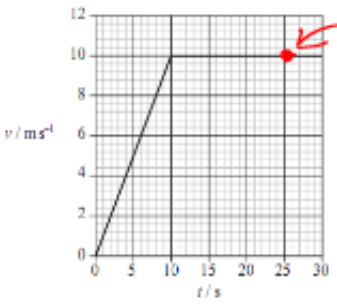
Uncertainty in I^2 = 2 × 10%

= 20%

(D)

2.

2.



25s

Speed = 10 ms^{-1}

average = $\frac{\text{dist}}{\text{time}}$

dist = area under graph

= 15 × 10 + ½ 10 × 10

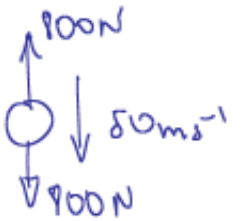
= 150 + 50 = 200m

Average speed = $\frac{200}{25}$ = 8 ms^{-1}

(C)

3.

3.



Const. speed + direction ⇒ forces balanced

(C)

4.

4. force is ⊥ to direction of motion
so UD = zero

(A)

5.

5. Uniform speed ⇒ Net force in direction of motion = zero
Circular motion ⇒ force ⊥ to direction of motion not balanced

(B)

6.

6. NO change in temp. ⇒ NO increase in KE
But heat is added so must be increase in PE

(A)

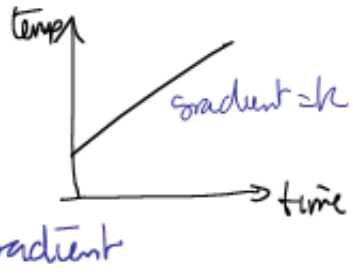
7.

7.

$Q = mc\Delta T$

$\frac{Q}{\Delta t} = mc\left(\frac{\Delta T}{\Delta t}\right)$ — gradient

$P = mck$



(A)

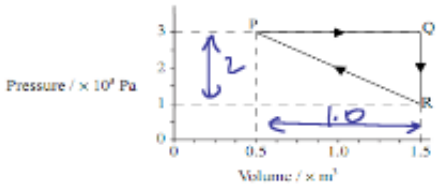
8.

8. First law of thermodynamics $Q = \Delta U + W$ is the same as cons. of Energy

(B)

9.

9.

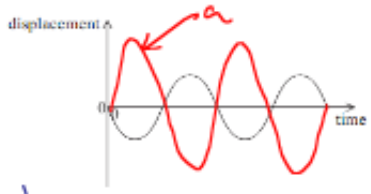


Net work = Area inside △
$= \frac{1}{2} \times 1.0 \times 2$
$= 1 \times 10^5 \text{ J}$

C

10.

10.

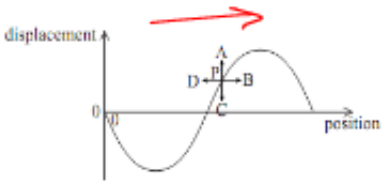


accel ∝ − x

D

11.

11.

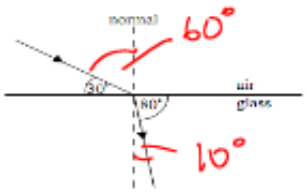


Wave travels → so particle is on its way down ↓

C

12.

12.



$n = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 10^\circ}$

C

13.

13.



$\Rightarrow 3 f_0 = 3 \times 330 = 990 Hz$

D

14.

14.



C

15.

15. If not resolved S_1 & S_2 are too close together $\Rightarrow$ $\theta < 1.22 \frac{\lambda}{b}$

A

16.

16. If $2 Wm^{-2}$ transmitted $\Rightarrow$ polarizer is at an angle to plane of polarization



$I = I_0 \cos^2\theta$

$\cos\theta = \sqrt{\frac{I}{I_0}} = \frac{1}{2}$

$\theta = 60°$

$I = I_0 \cos^2(\theta - 90°)$

$= I_0 \cos^2(30)$

$= I_0 \times 0.25$

$6 Wm^{-2}$

Don't need to do calculation can eliminate 0, 2, and 8 $\Rightarrow$ 6

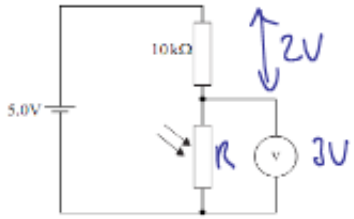
C

17.

17. $1eV = 1.6 \times 10^{-19} J$
simple as that (B)

18.

18. - Swapping LDR and 10kΩ will give pd = 2V ✗



- Increase 10kΩ will reduce current → reduced pd across LDR ✗

- Increasing light on LDR → reduces R → more current → increased pd across 10kΩ → reduced pd across LDR ✗
must be the next one

Decreasing light on LDR → reduces R etc. ✓

(D)

19.

19.



$\mathcal{E} = \frac{d N\Phi}{dt}$ rate of change of flux

Hmm? is it linkage or through?
Sound almost the same but I'll go for Linkage (more technical sounding) (A)

20.

20. Instantaneous Power = $I_{peak} \times V_{peak}$

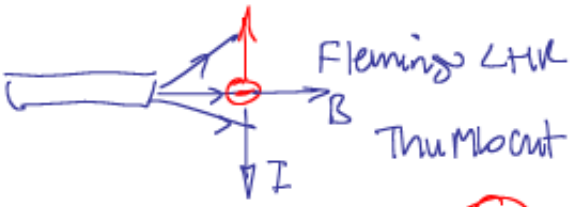
$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$

$\Rightarrow I_{peak} V_{peak} = \sqrt{2} I_{rms} \sqrt{2} V_{rms}$

$= 2VI$ (A)

21.

21.



(B)

22.

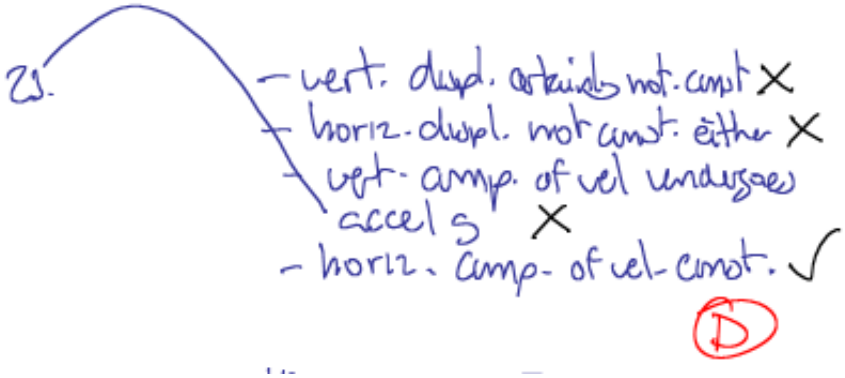
22 Electric field? Bit of a trick question!

Force must always be directed towards centre +ve charge ⇒ accelerated in direction of field



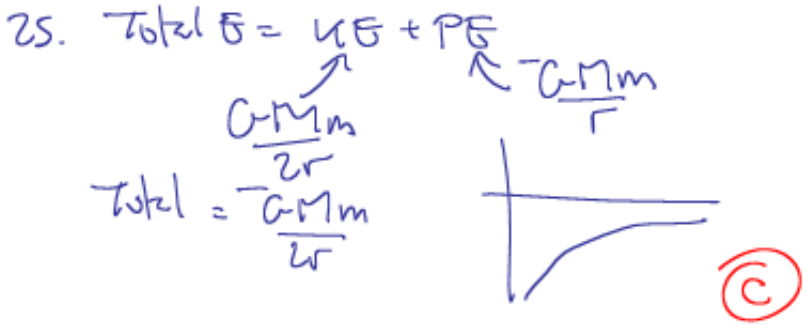
(A)

23.

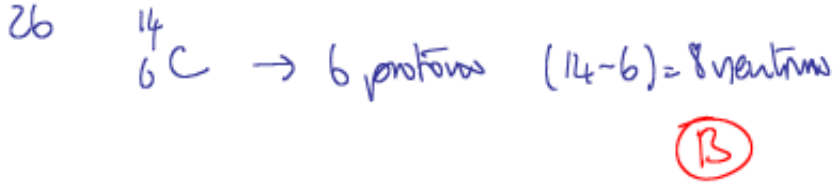

23.
- vert. displ. at t=0 not const ✗
- horiz. displ. not const. either ✗
- vert. comp. of vel undergoes accel g ✗
- horiz. comp. of vel const. ✓
D

24.

25.


25. Total E = KE + PE
KE = GMm/2r
PE = -GMm/r
Total = -GMm/2r
C

26.

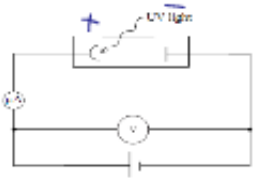

26. 14 6C → 6 protons (14-6) = 8 neutrons
B

27.


27. Alpha is most ionising
A

28.

28.



If current stops then no electrons travel from cathode → anode.

– PD is opposing e^- motion so increasing it will stop the e^-s ✓

– Increasing f, if we assume I remains const then each photon has less energy ⇒ less photons ⇒ less current. Hmm! well maybe number of photons is constant then I would increase, e^- will have more energy → current will increase. Either way the current won't STOP ✗

– Increase intensity ⇒ more photons ⇒ more electrons ⇒ more current ✗

– smaller ∅ ⇒ electrons will have higher KE ⇒ greater current. ✗

(A)

29.

29. $\frac{1}{2}mv^2 = Ve \Rightarrow v \propto \sqrt{V}$

so if V×2 v increased × $\sqrt{2}$

⇒ momentum × $\sqrt{2}$

$\lambda = \frac{h}{p}$ so if $p \times \sqrt{2}$

$\lambda \times 1/\sqrt{2}$

(C)

30.

30. Questions says UE so not $\equiv$ $\equiv$ which gives the 6level

$UE = \frac{n^2 h^2}{8mL^2}$ so $UE \propto n^2$

⇒ as n increases the levels get further apart.

$\equiv$ $\equiv$ (C)

31.

31. Heisenberg — $\Delta p \Delta x \geq \frac{h}{4\pi}$

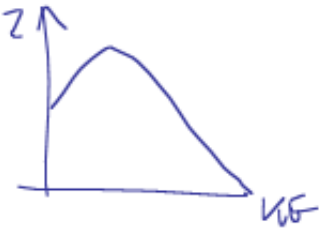
Δx must be less than size of nucleus $\sim 10^{-15}$m

$\Rightarrow \Delta p \sim \frac{h}{10^{-15}} = \frac{10^{-34}}{10^{-15}} = 10^{-19}$

$\sim 10^{-20}$ Ns (B)

32.

32. β have continuous energy spectrum



(B)

33.

33.

After 10 – 50%
20 – 25% ← between these two
30 – 12.5%

18%?

(B)

34.

34 Generator converts mechanical → electrical energy

D

35.

35 $P = \frac{1}{2}\rho\pi r^2 v^3$

$P \propto v^3$ so if $2v \Rightarrow 8 \times P$

D

36.

36. IR resonates with CO_2 molecules

A

37.

37. Uranium has most energy/kg provided it has been processed

D

38.

38.

16	8	4	2	1	
1	0	0	1	0	= 16 + 2 = 18
↑				↑	
MSB				LSB	

D

39.

39.



for destructive int path diff = $\lambda/2$

$pd = 2d = \frac{\lambda}{2}$ $\lambda = 4d$

D

40.

40. Well my watch is my mobile which is also my camera and this does have a CCD in it. But I guess that watch is the answer.

C

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